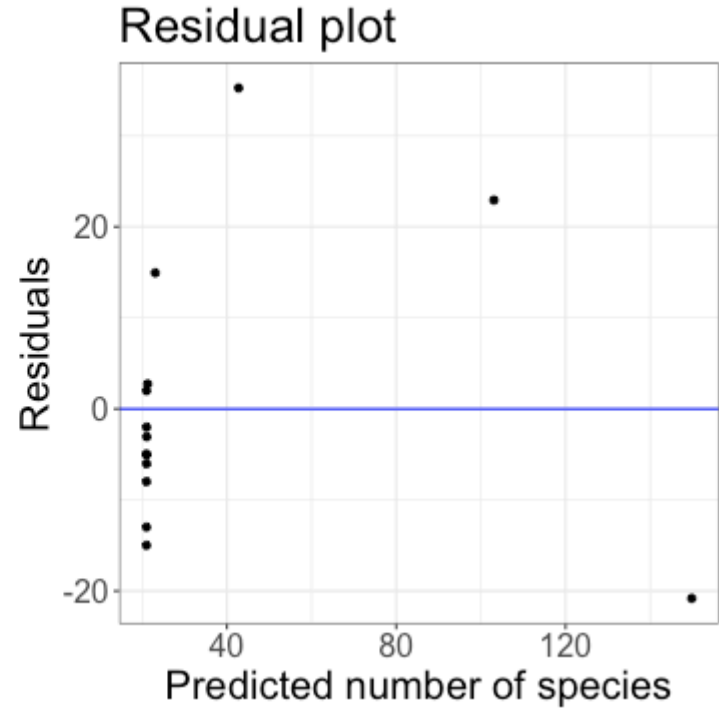
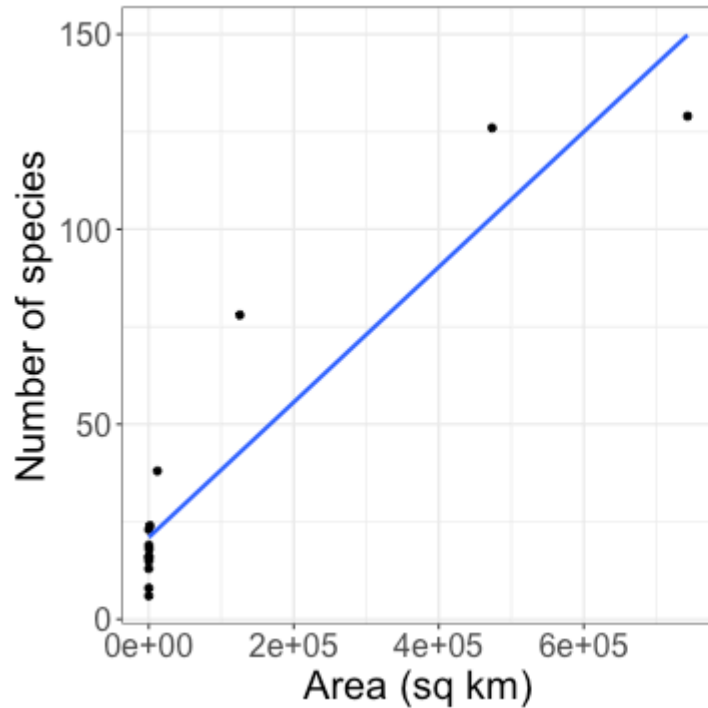


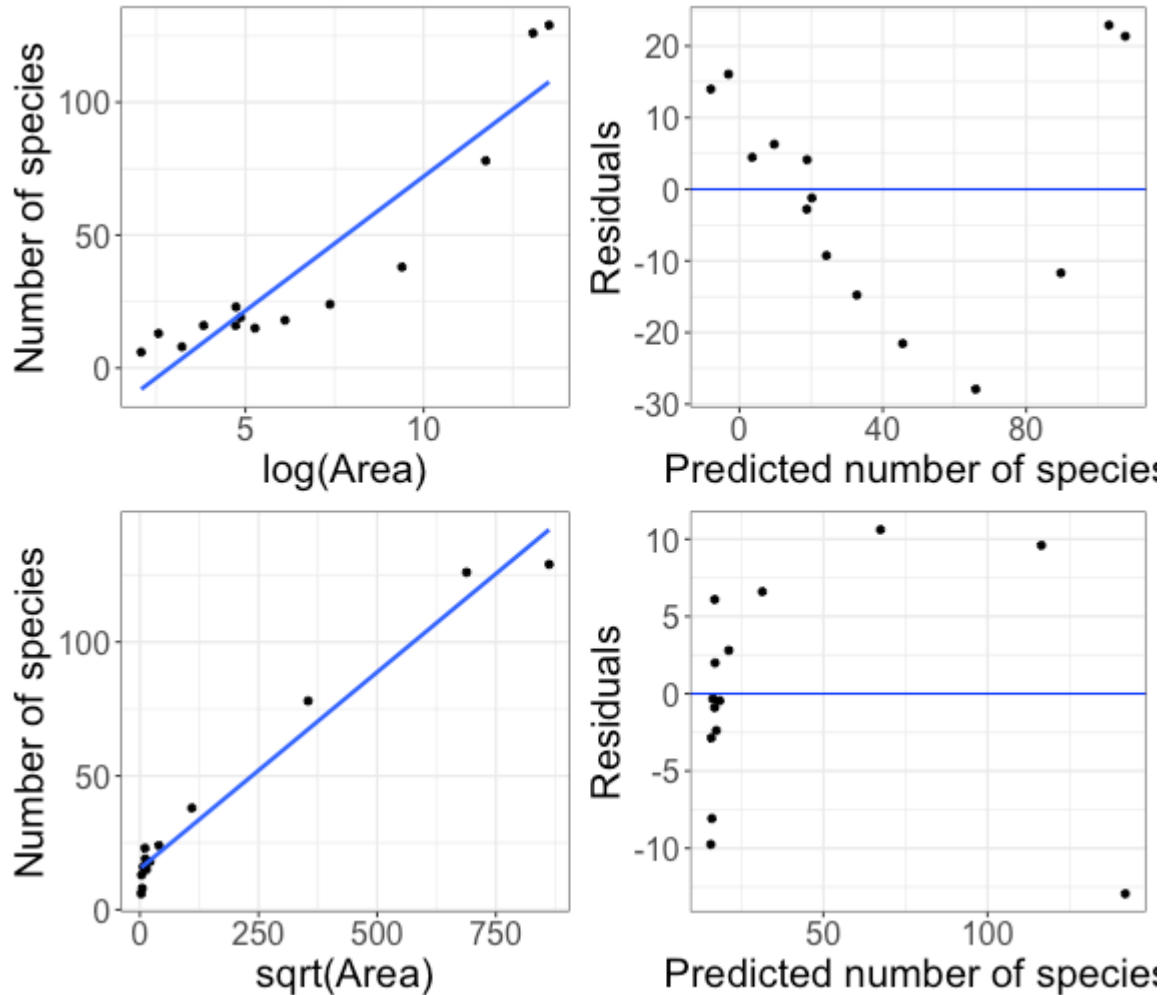
Nonlinear relationships and transformations

More transformations



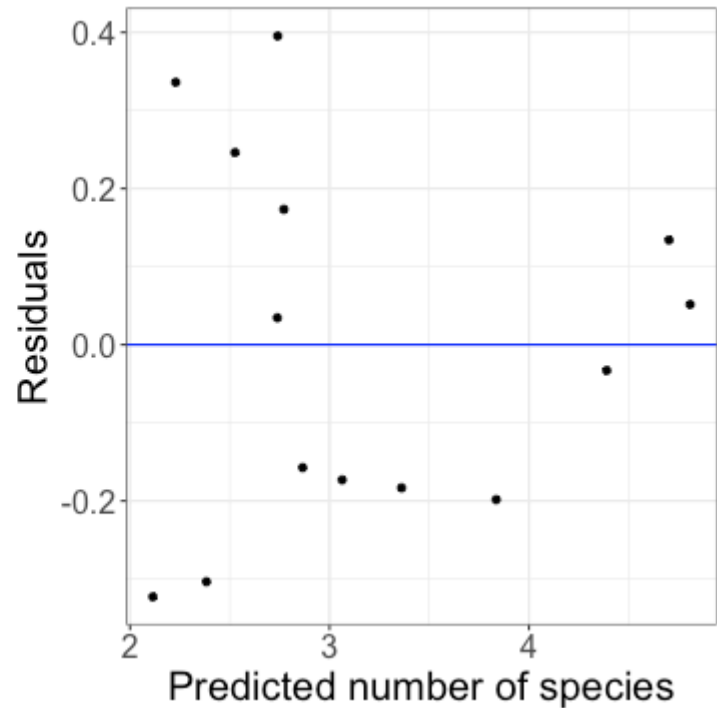
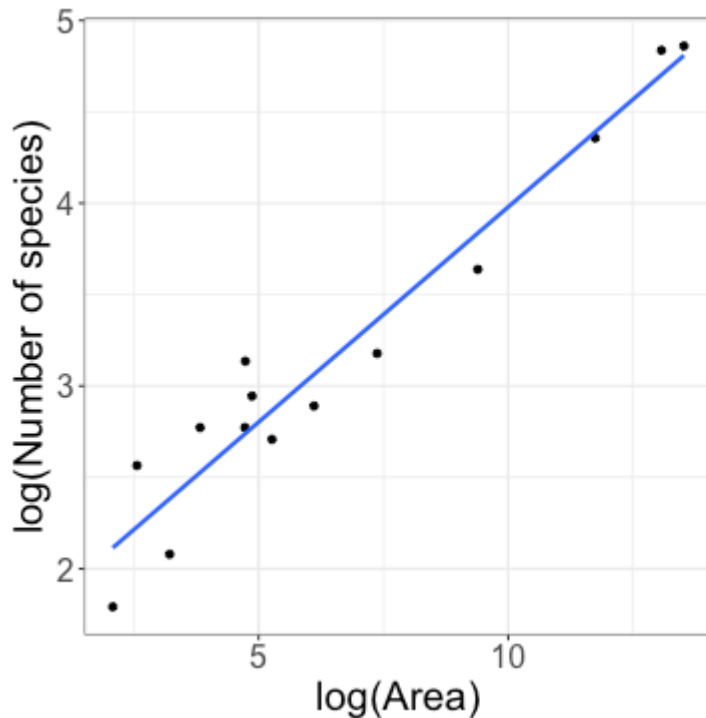
What transformations could we try here?

Trying some transformations



Transform both predictor and response?

$$\log(\textit{Species}) = \beta_0 + \beta_1 \log(\textit{Area}) + \varepsilon$$



Interpreting the fitted model

$$\log(\widehat{Species}) = 1.625 + 0.246 \log(Area)$$

Summary of model interpretation so far

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

- + A one-unit increase in x is associated with a change in y of $\hat{\beta}_1$, on average

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 \log(x)$$

- + A 1% increase in x is associated with a change in y of $\hat{\beta}_1/100$, on average

$$\widehat{\log(y)} = \hat{\beta}_0 + \hat{\beta}_1 x$$

- + A one-unit increase in x is associated with a change in y by a **factor** of $e^{\hat{\beta}_1}$, on average

$$\widehat{\log(y)} = \hat{\beta}_0 + \hat{\beta}_1 \log(x)$$

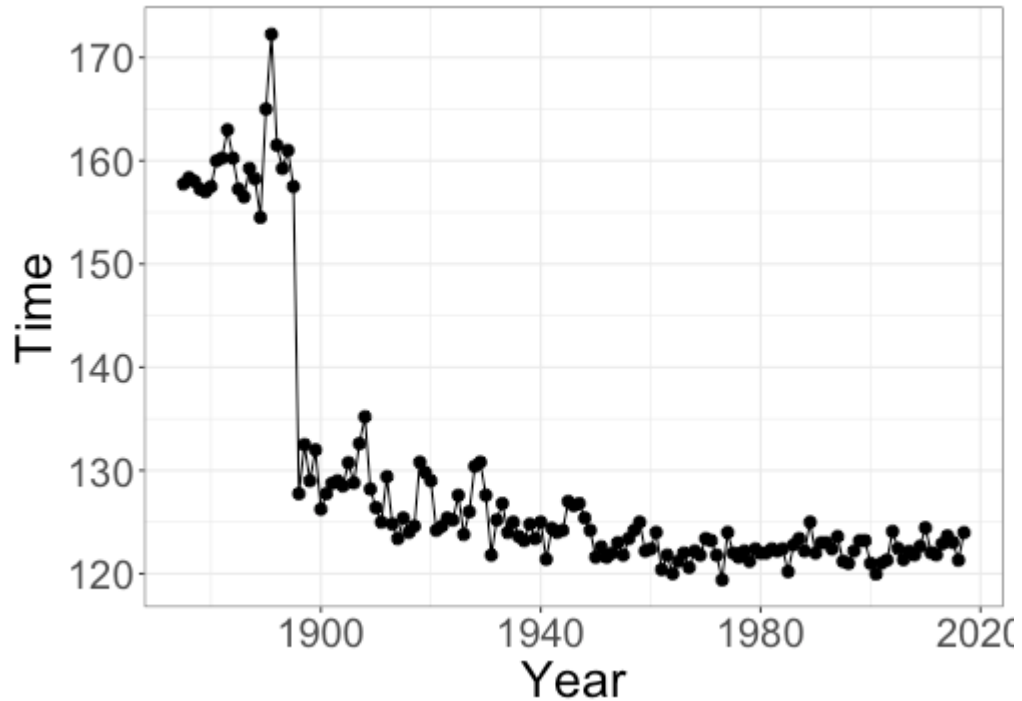
- + A 1% increase in x is associated with a change in y by a **factor** of $(1.01)^{\hat{\beta}_1}$, on average

Kentucky Derby data

Data on the Kentucky Derby winner from each year between 1875 and 2012. Variables include

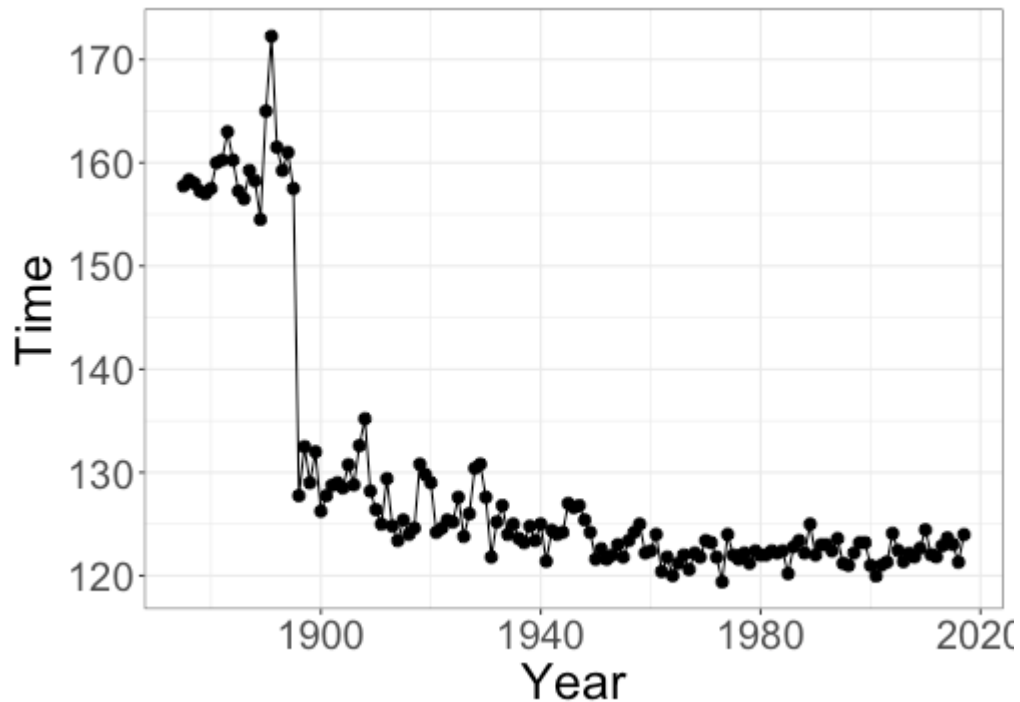
- + year: year of race
- + speed: speed of winning horse (mph)
- + time: winning time (seconds)
- + condition: condition of the track on the day of the race

EDA



What do you notice about the winning times?

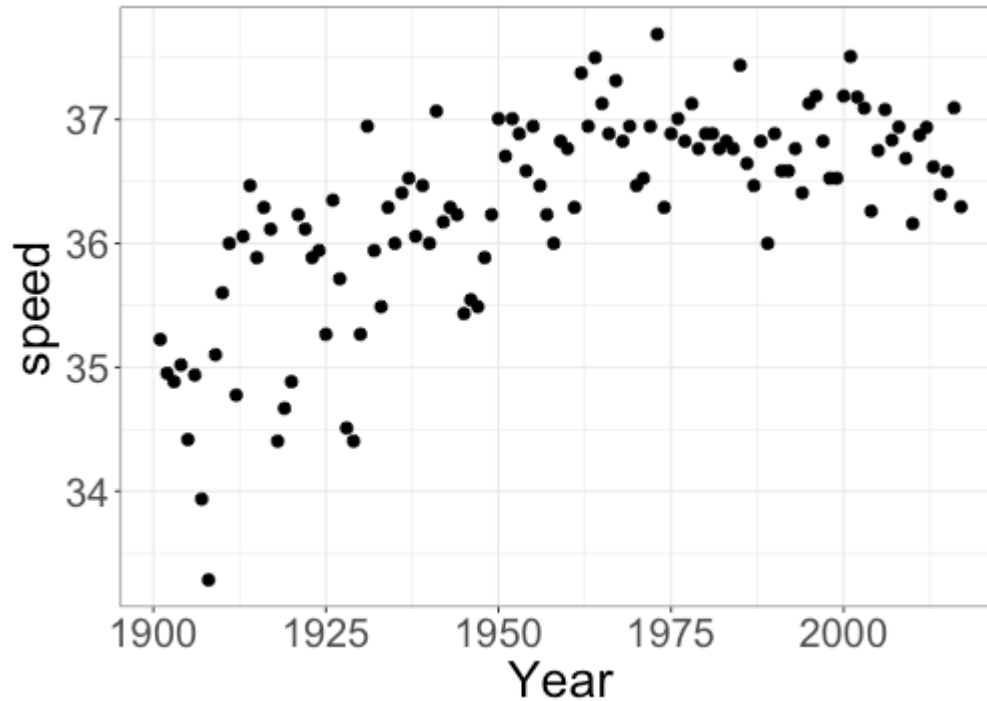
EDA



What do you notice about the winning times?

There's a big drop around 1900, then they decrease gradually.

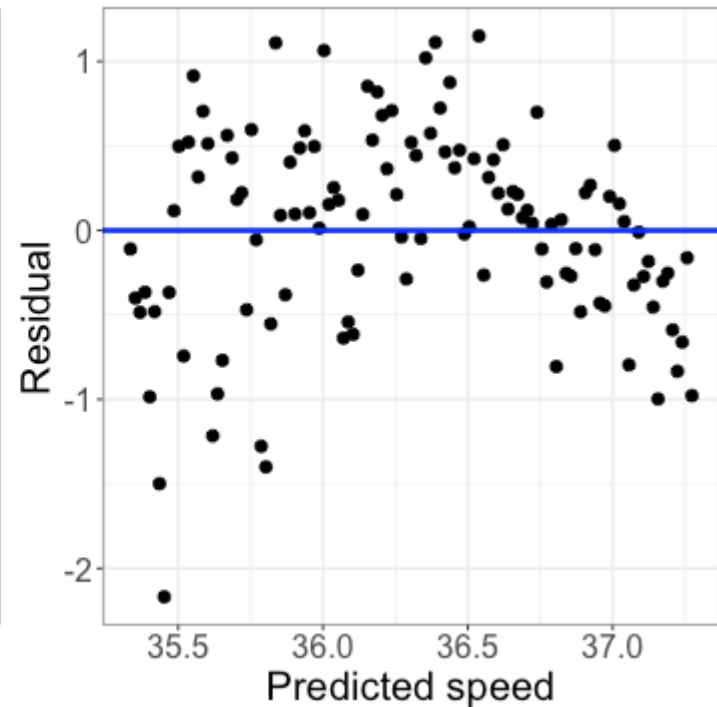
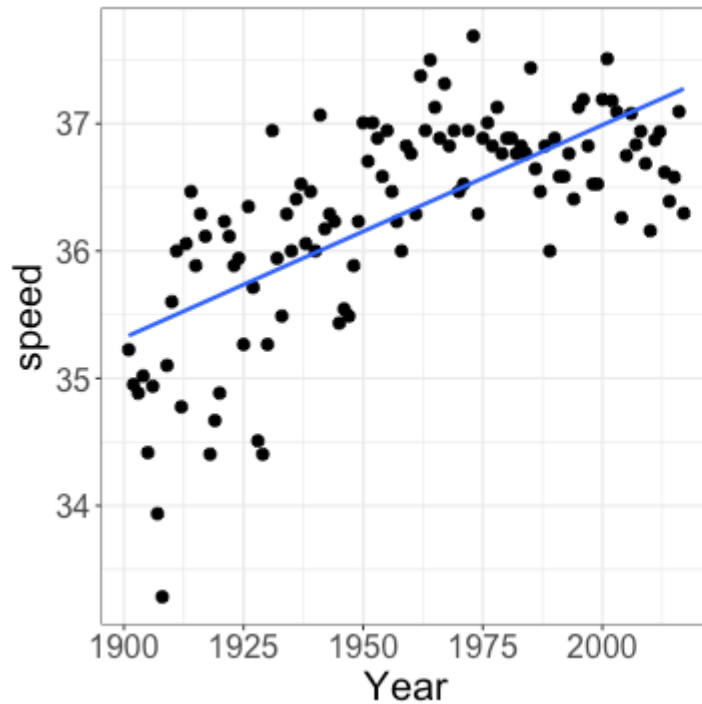
EDA



How would you describe the relationship?

Fitting a linear regression

$$\text{Speed} = \beta_0 + \beta_1 \text{Year} + \varepsilon$$



Polynomial transformations

Another possible transformation is a **polynomial**.

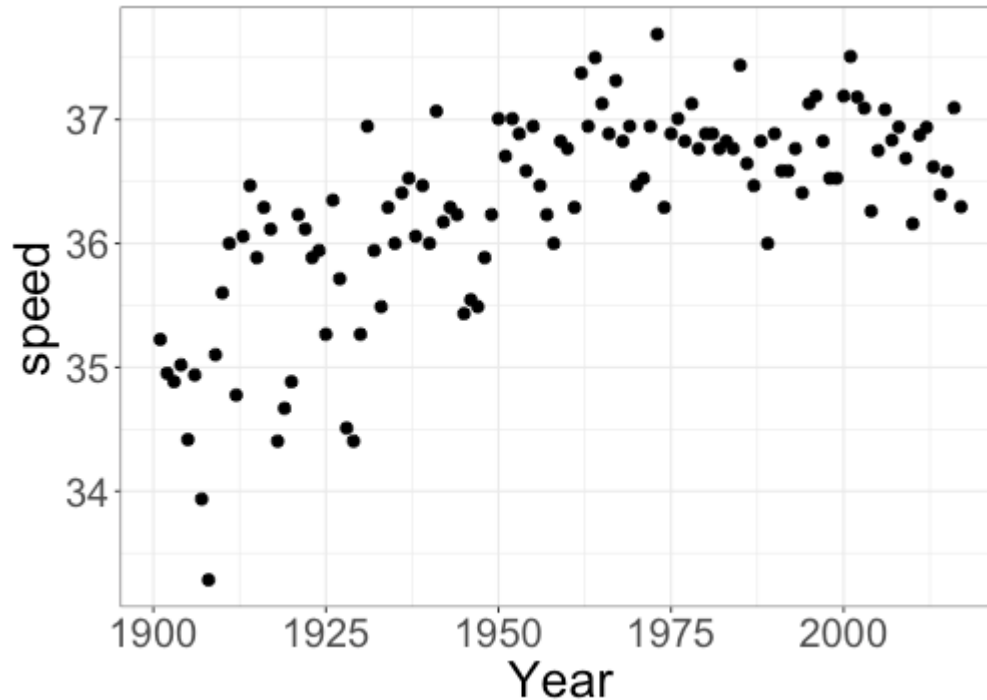
Definition: A polynomial transformation of order k is a linear combination of the terms $X, X^2, \dots, X^k$:

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \dots + \beta_k X^k + \varepsilon$$

+ Example: If we use a polynomial of order 3, our model is

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \varepsilon$$

Polynomial transformations

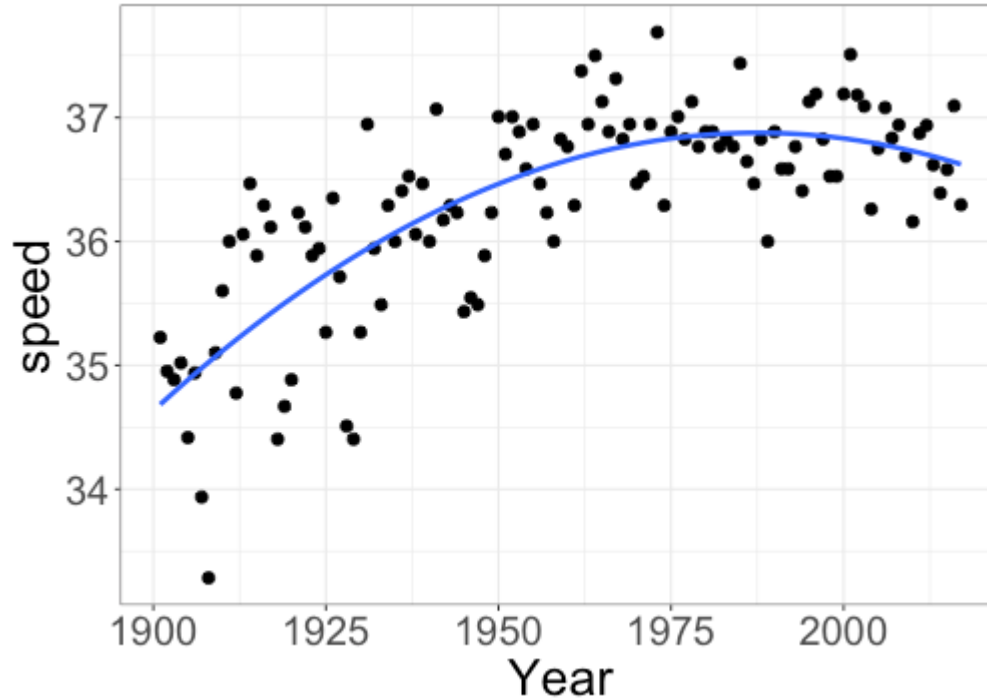


What order polynomial do you think we should use?

Polynomial transformation

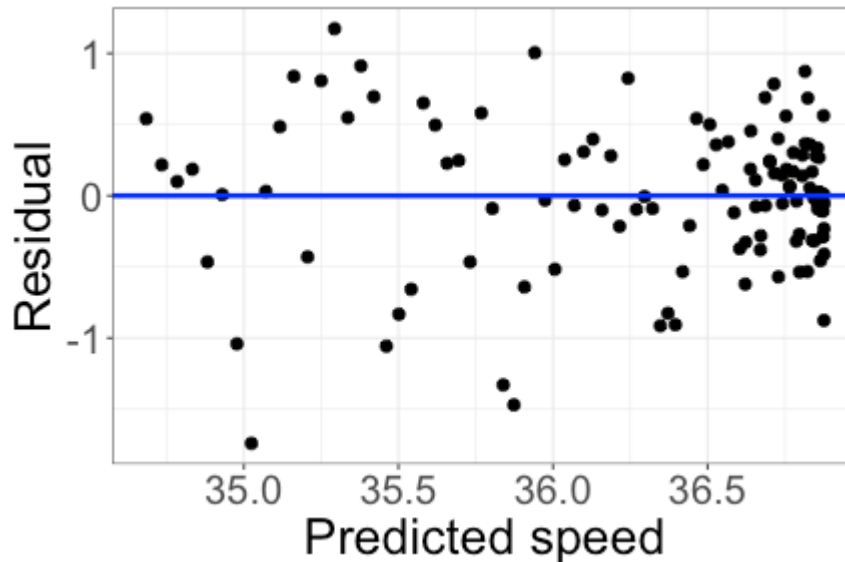
Let's try a polynomial of order 2:

$$\text{Speed} = \beta_0 + \beta_1 \text{ Year} + \beta_2 \text{ Year}^2 + \varepsilon$$



Diagnostic plots

We use residual plots to check the model assumptions after transforming our predictor, just like with other transformations.



How does the shape assumption look now?

Fitting the model in R

```
derby_lm <- lm(speed ~ poly(Year, 2, raw=TRUE),  
               data = derby)
```

- + `poly` means "fit a polynomial"
- + `poly(..., 2, ...)` specifies a second-order polynomial (quadratic)

Fitting the model in R

```
derby_lm <- lm(speed ~ poly(Year, 2, raw=TRUE),  
               data = derby)  
derby_lm
```

```
##  
## Call:  
## lm(formula = speed ~ poly(Year, 2, raw = TRUE), data = d  
##  
## Coefficients:  
##              (Intercept)  poly(Year, 2, raw = TRUE)1  
##              -1.120e+03                1.164e+00  
## poly(Year, 2, raw = TRUE)2  
##              -2.928e-04
```

$$\widehat{\text{Speed}} = -1120 + 1.16 \text{ Year} - 0.00029 \text{ Year}^2$$

Interpreting the fitted model

$$\widehat{\text{Speed}} = -1120 + 1.16 \text{ Year} - 0.00029 \text{ Year}^2$$

How would I interpret the intercept of the fitted polynomial regression model?

Interpreting the fitted model

$$\widehat{\text{Speed}} = -1120 + 1.16 \text{ Year} - 0.00029 \text{ Year}^2$$

How would I interpret the intercept of the fitted polynomial regression model?

If Year = 0, we predict a speed of -1120 mph (doesn't make much sense, since year is nowhere near 0)

Interpreting the fitted model

$$\widehat{\text{Speed}} = -1120 + 1.16 \text{ Year} - 0.00029 \text{ Year}^2$$

How would I interpret the terms on Year and Year²?

Interpreting the fitted model

$$\widehat{\text{Speed}} = -1120 + 1.16 \text{ Year} - 0.00029 \text{ Year}^2$$

How would I interpret the terms on Year and Year²?

They're kind of difficult to interpret -- if I change Year, I also change Year²

Interpreting the fitted model

$$\widehat{\text{Speed}} = -1120 + 1.16 \text{ Year} - 0.00029 \text{ Year}^2$$

Can I interpret the *sign* on Year^2 ?

Interpreting the fitted model

$$\widehat{\text{Speed}} = -1120 + 1.16 \text{ Year} - 0.00029 \text{ Year}^2$$

Can I interpret the *sign* on Year^2 ?

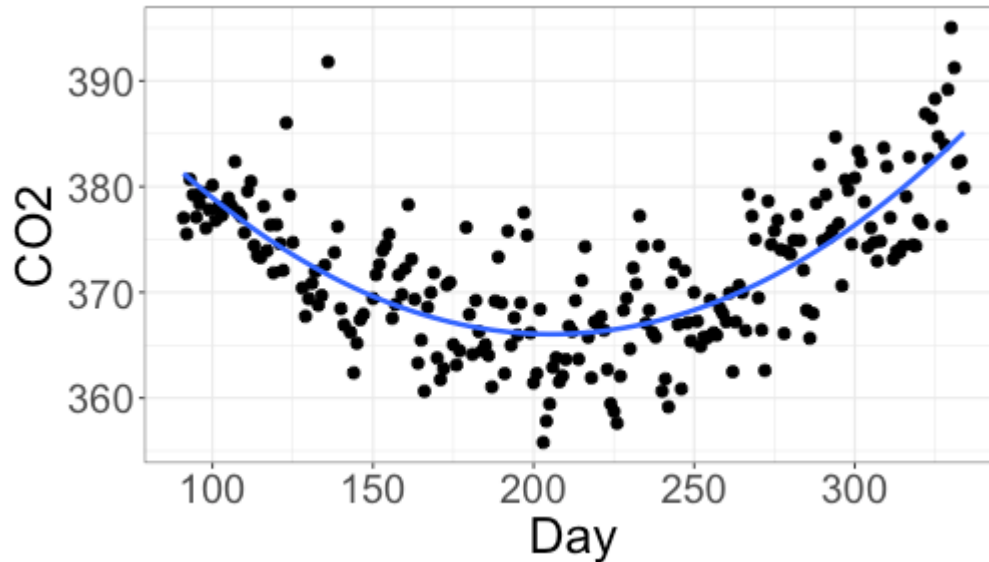
Yes (for a polynomial of order 2 specifically). The sign on Year^2 tells us about the concavity of the relationship.

Class activity

https://sta112-s26.github.io/class_activities/ca_10.html

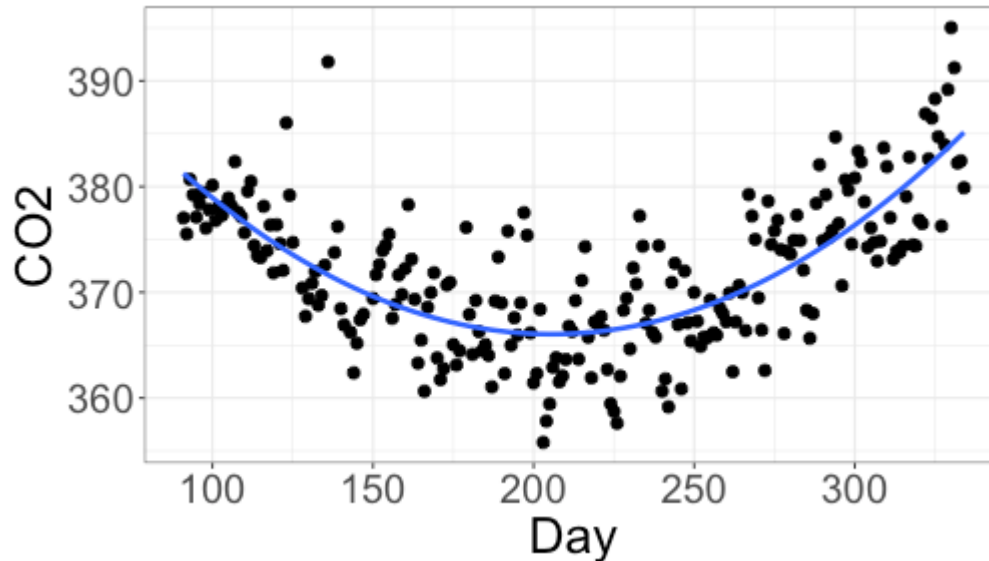
- + Experiment with polynomial regression
- + Submit HTML at end of class

Previously: polynomial transformations



$$\widehat{\text{CO2}} = 414.975 - 0.476 \text{ Day} + 0.0012 \text{ Day}^2$$

Previously: polynomial transformations



$$\widehat{\text{CO2}} = 414.975 - 0.476 \text{ Day} + 0.0012 \text{ Day}^2$$

Can I use the correlation to calculate the relationship between Day and CO2?

Measuring variability in the response

- + When we fit a regression model with least squares, we choose parameters to minimize the sum of squared residuals:

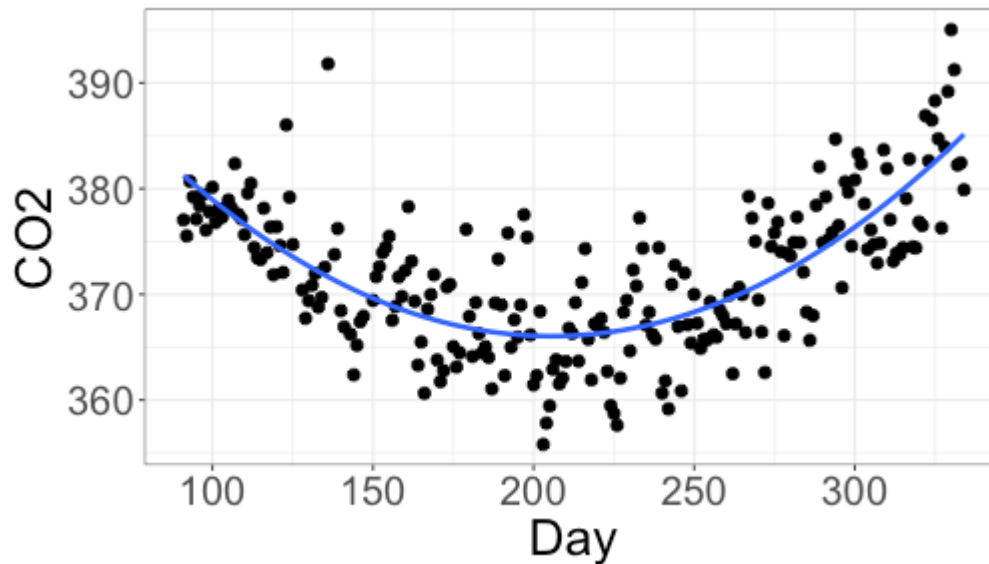
$$SSE = \sum_i (y_i - \hat{y}_i)^2$$

- + We can think of SSE as the "amount of variability" in Y after predicting with our regression model
- + The amount of variability in Y *before* predicting with our regression model is

$$SSTotal = \sum_i (y_i - \bar{y})^2$$

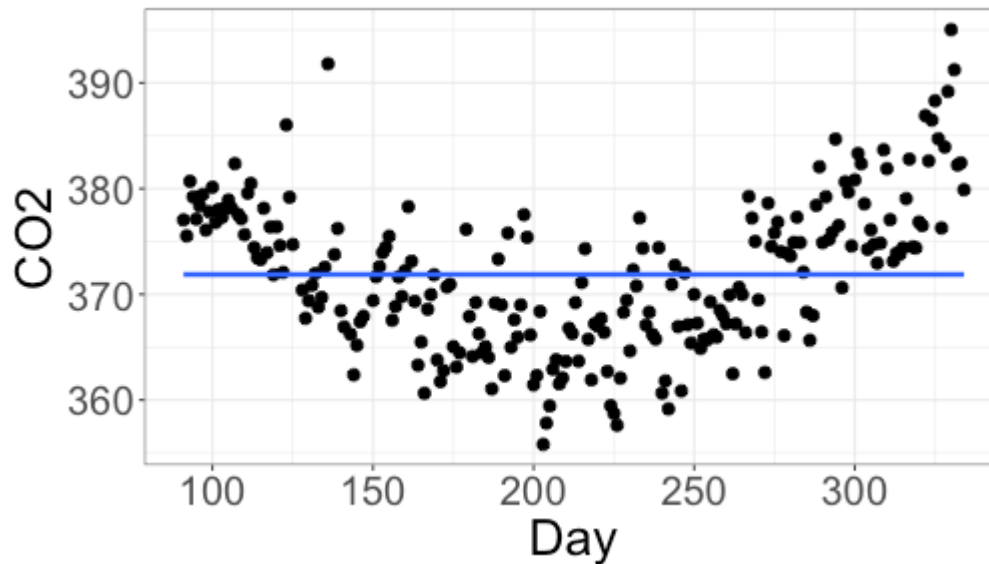
Thinking about SSE

$$SSE = \sum_i (y_i - \hat{y}_i)^2$$

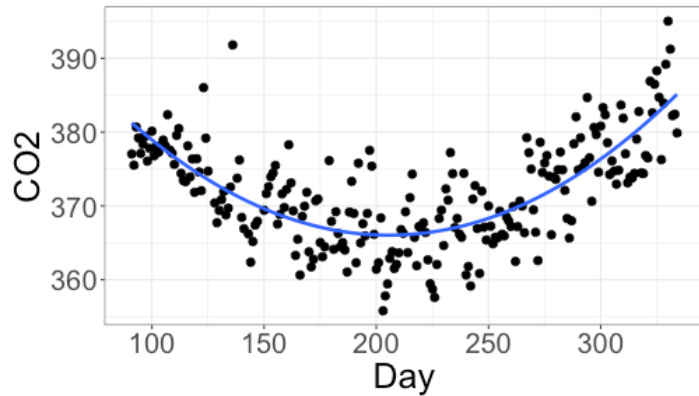


Thinking about SS_{Total}

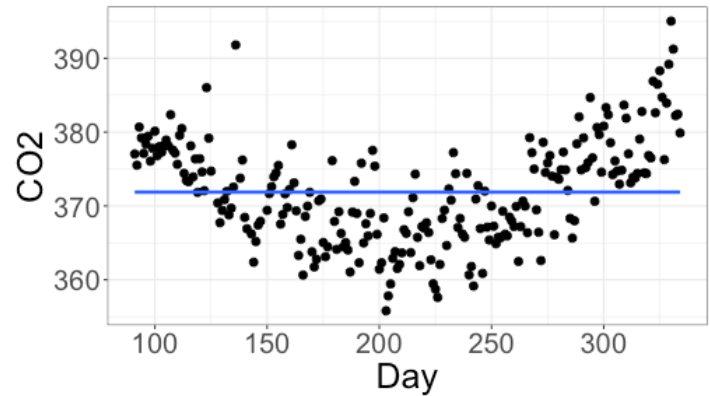
$$SS_{Total} = \sum_i (y_i - \bar{y})^2$$



SSE vs. SSTotal



SSE = 4992.71



SSTotal: 11703.69

Do I want SSE to be large relative to SSTotal, or small relative to SSTotal?

R^2 : the coefficient of determination

$$SSE = \sum_i (y_i - \hat{y}_i)^2$$

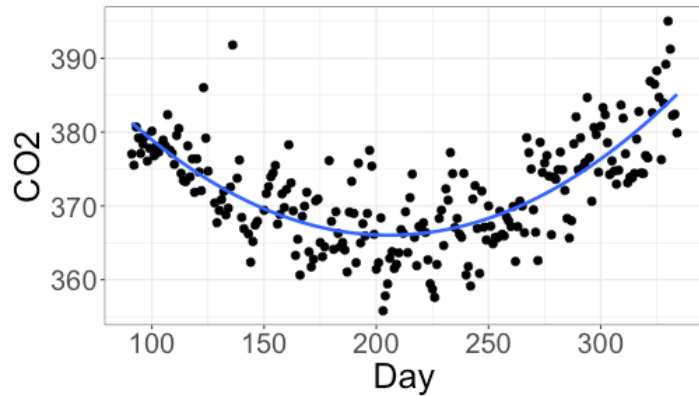
$$SSTotal = \sum_i (y_i - \bar{y})^2$$

The proportion of variability in Y "explained" by our regression model is

$$R^2 = 1 - \frac{SSE}{SSTotal}$$

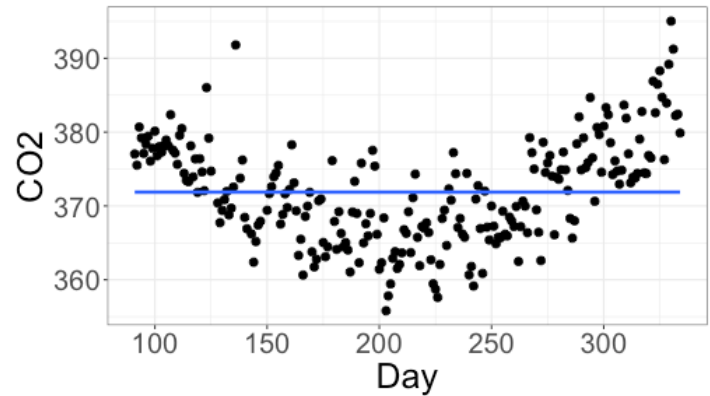
- + R^2 is between 0 and 1
- + A higher R^2 means our model is a "better fit" to the data
- + For regression with one variable, $R^2 = r^2$ (the squared correlation)

SSE vs. SSTotal



SSE = 4992.71

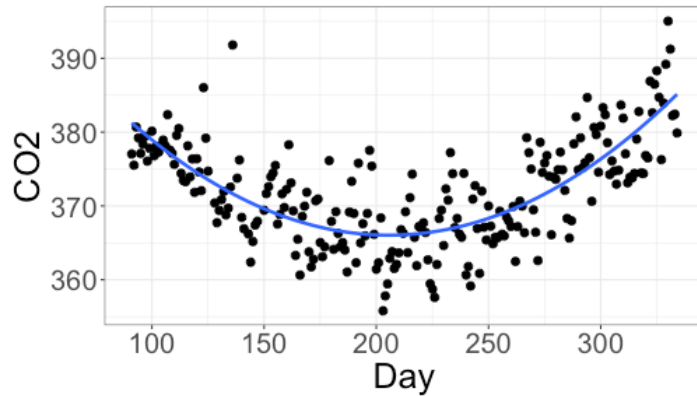
$$R^2 = 1 - \frac{4992.71}{11703.69} = 0.573$$



SSTotal: 11703.69

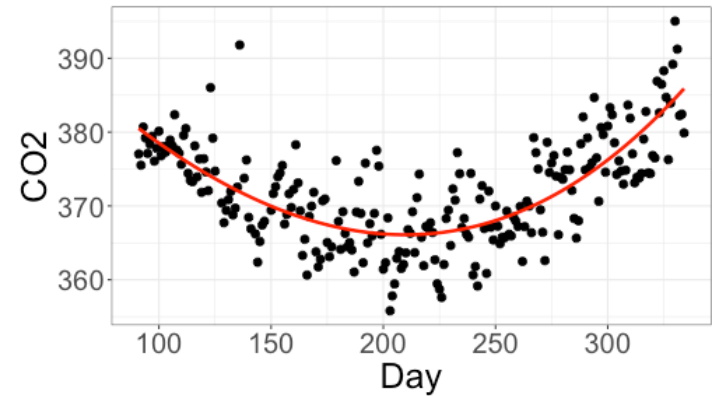
Comparing R^2 values

Polynomial of order 2:



$$R^2 = 0.573$$

Polynomial of order 3:



$$R^2 = 0.575$$

R^2 has increased very slightly -- does that mean that a polynomial of order 3 is a better fit?

R^2

R^2 will always increase when you add more parameters to the model.

Why? Because

$$R^2 = 1 - \frac{SSE}{SSTotal}$$

and adding more parameters gives us more flexibility to decrease SSE.

Adjusted R^2

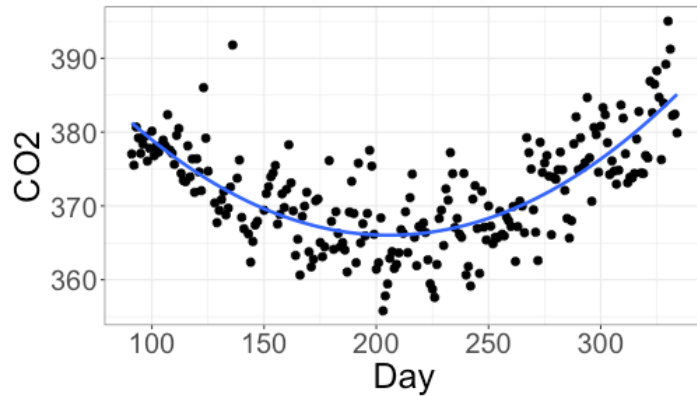
We can account for the number of parameters in the model with *adjusted R^2* :

$$R_{adj}^2 = 1 - \frac{SSE/(n - p)}{SSTotal/(n - 1)}$$

- + n = number of observations in data (sample size)
- + p = number of parameters (β terms) in model

Adjusted R^2

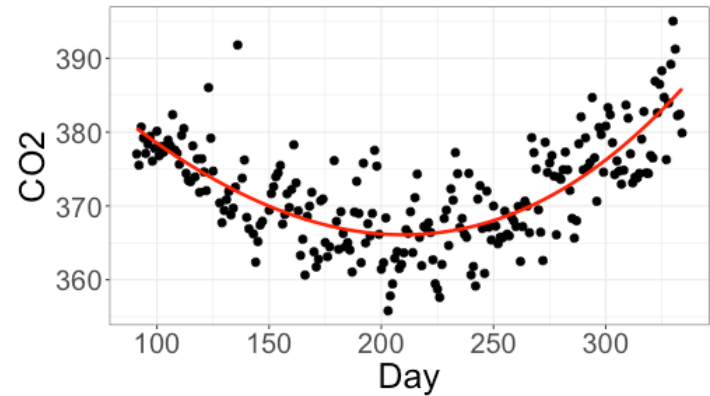
Polynomial of order 2:



SSE = 4992.7

SSTotal = 11703.69

Polynomial of order 3:



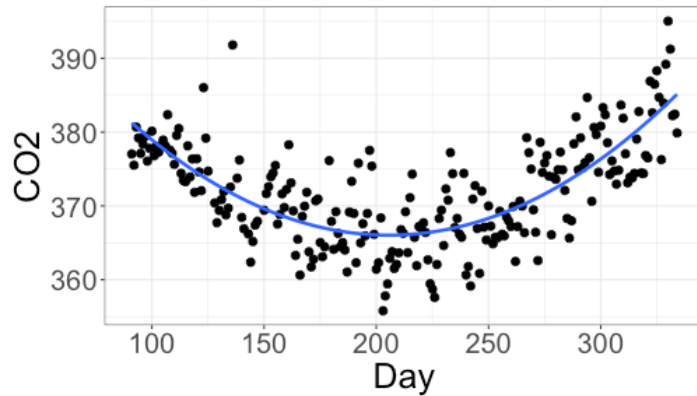
SSE = 4971.9

SSTotal = 11703.69

Why is SSTotal the same for the two models?

Adjusted R^2

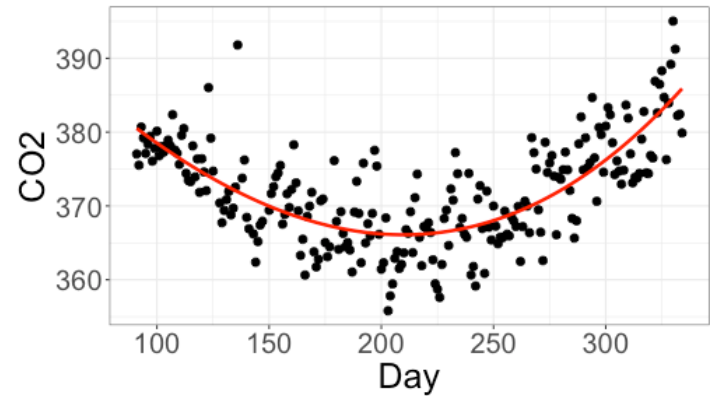
Polynomial of order 2:



SSE = 4992.7

SSTotal = 11703.69

Polynomial of order 3:



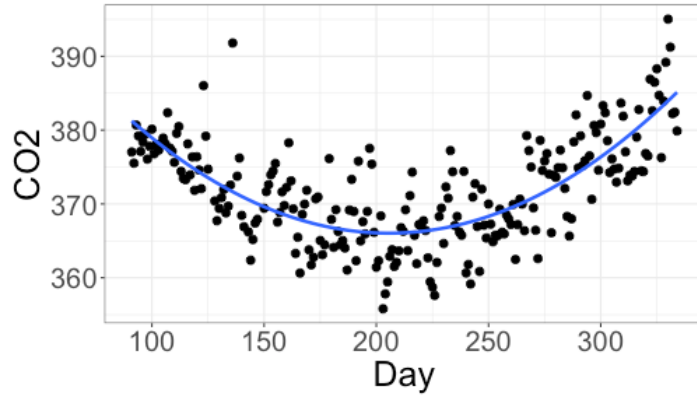
SSE = 4971.9

SSTotal = 11703.69

$$R_{adj}^2 = 1 - \frac{SSE/(n - p)}{SSTotal/(n - 1)}$$

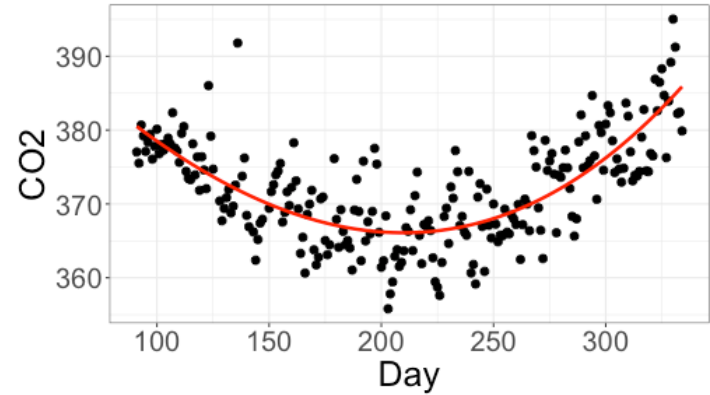
Adjusted R^2

Polynomial of order 2:



$$R_{adj}^2 = 0.56976$$

Polynomial of order 3:



$$R_{adj}^2 = 0.56972$$